

Abhishek Pawar

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Summary

AI/ML Engineer with 1.5+ years of experience designing and shipping production multi-agent LLM systems on Google ADK, Model Context Protocol (MCP), and Gemini/OpenAI/DeepSeek APIs for regulated clinical-data platforms. Delivered multi-agent orchestration (6+ topologies), natural-language-to-SQL pipelines with defense-in-depth guardrails, and event-driven data infrastructure (Kafka, Airflow) with full-stack delivery on FastAPI, React, PostgreSQL, and MongoDB — deployed on Azure with CI/CD. Strong foundation in Python, distributed systems, and reliability engineering for LLM-powered applications.

Education

Jawaharlal Nehru Engineering College, MGM's University

May 2025

Bachelor of Technology in Computer Science and Engineering (CGPA: 9.07/10.00)

Aurangabad, Maharashtra

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java), Database Management Systems (DBMS), Operating Systems, Computer Networking, DevOps

Experience

Dizzaroo Pvt Ltd

Feb 2025 – Present

AI/ML Engineer

Pune, Maharashtra

- Architected 6 distinct multi-agent orchestration topologies (graph, coordinator-delegate, hierarchical, iterative-refinement, sequential, parallel) on Google Agent Development Kit (ADK), compiling user-authored agent configurations into live agent execution trees at runtime.
- Built a no-code multi-agent platform enabling non-technical users to visually design agent workflows (@xyflow/react drag-and-drop canvas) and configure each agent's model, tools, and instructions through the UI; backend dynamically compiled these MongoDB-stored configurations into live, runnable Google ADK agent trees at request time – eliminating the need to hand-write agent code per workflow.
- Self-hosted Model Context Protocol (MCP) servers and clients exposing 29+ tools across PostgreSQL, Neo4j, MongoDB, and Weaviate, plus 6 external clinical, regulatory, and financial data APIs, with a global retry plugin that automatically recovered failed tool calls up to 2x without terminating the run.
- Engineered a natural-language-to-SQL query and dashboard generation system with a defense-in-depth guardrail layer (sqlglot static validation, read-only allow-list parsing, per-tenant data-access policy), verified by 14 dedicated security unit tests blocking SQL injection and destructive statements.
- Integrated 3 LLM providers (OpenAI, Google Gemini, DeepSeek) with a dedicated safety-classifier gate (IBM Watsonx Granite Guardian) screening every generation for jailbreak, harm, and groundedness risk.
- Designed per-task, cost-tiered LLM model routing across 15+ configuration points, tiering cheaper models for high-volume classification against stronger models for synthesis and planning.
- Built a per-study document library feature enabling users to upload and update source files, with automated data extraction cross-referenced against live web search and a PubMed MCP server, surfacing mismatches between extracted and referenced data for human-in-the-loop review and approval before updates were finalized.
- Orchestrated event-driven data infrastructure: an idempotent Kafka consumer/producer pipeline for cross-service entity synchronization, and Apache Airflow ETL pipelines converting raw EDC data into SDTM-compliant PostgreSQL datasets with automatic rollback on failure.
- Shipped real-time streaming (Server-Sent Events + WebSocket) with exponential-backoff reconnect logic and a visual drag-and-drop multi-agent workflow builder (@xyflow/react); deployed full-stack applications on Azure App Service, Azure Static Web Apps, and Azure Container Registry via Azure Pipelines CI/CD.

Tech Mahindra

Dec 2024 – Feb 2025

AI Intern

Pune, Maharashtra

- Designed and developed a no-code multi-agent platform using the CrewAI framework, enabling users to create agents, define tasks, and wire tool integrations through a visual interface without writing code.
- Architected RESTful APIs and MongoDB schemas for core entities (Agent, Task, Tool, User) to support dynamic agent orchestration, enabling modular and extensible multi-agent workflow execution.

Projects

Smart Finance Manager | *Next.js, PostgreSQL, Shadcn*

- Developed a full-featured finance management platform with support for multiple account types, custom categories, and CSV import for bulk transaction uploads simplifying financial tracking for both individuals and small businesses.
- Designed an interactive analytics dashboard with various chart types to visualize spending patterns, integrated Clerk for secure user authentication, and used Docker to ensure consistent and efficient deployment across environments.

Clinical RAG Assistant | *React, Weaviate, Python, FastAPI*

- Created a retrieval-augmented generation assistant for medical and clinical documents, implementing paragraph-based chunking and hybrid search over a Weaviate vector database for accurate, context-aware retrieval.
- Combined BM25 sparse retrieval with dense embeddings in a hybrid search strategy to surface both keyword-exact and semantically relevant chunks for LLM response generation.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, HTML/CSS

AI & Agents: Google ADK, Multi-Agent Orchestration (Graph, Hierarchical, Sequential, Parallel, Coordinator, Iterative-Refinement), Model Context Protocol (MCP), OpenAI, Google Gemini, DeepSeek, Prompt Engineering, Structured Output (Pydantic), RAG, Grounding, Guardrails, IBM Watsonx Granite Guardian

Retrieval & Embeddings: Weaviate, Hybrid Search (BM25 + Dense), Vector Search

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Swagger (OpenAPI), WebSocket, Server-Sent Events (SSE), Pydantic, SQLGlott

Databases & Messaging: PostgreSQL, MongoDB, Neo4j, Apache Kafka, Apache Airflow

Frontend: React 19, Vite, Zustand, TypeScript, TanStack Table, @xyflow/react

Cloud & DevOps: Azure App Service, Azure Static Web Apps, Azure Container Registry, Azure Virtual Machine, Azure Blob Storage, Docker, Azure Pipelines (CI/CD), Git

Observability: OpenTelemetry, MLflow, AgentOps, Structured Logging

Developer Tools: VS Code, Cursor, Claude Code, Postman, GitHub

Operating Systems: Ubuntu, Rocky Linux, OpenSUSE

Certifications and Achievements

- **Certified Linux Administrator (CLA)** – Linux Academy Australia
- **Model Context Protocol: Advanced Topics** – Anthropic Academy
- **AWS Cloud Practitioner Essentials** – Amazon Web Services (AWS)
- **Deep Research with LangGraph** – LangChain